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MINOR PROJECT FINAL REPORT

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WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar

Submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Technology in Computer Science & Engineering

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING • ACADEMIC SESSION 2023–2027
(OCTOBER 2026)

CANDIDATE'S DECLARATION

Amity School of Engineering & Technology • Amity University Uttar Pradesh

I, **Anubhav Anand** (Enrollment No. **A41105223039**), student of **Bachelor of Technology in Computer Science & Engineering**, Batch **2023–2027 (Semester VII)**, Amity School of Engineering & Technology (ASET), Amity University Uttar Pradesh, Greater Noida, hereby declare that the Minor Project report titled *"WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar"* represents original software engineering, data engineering, and statistical auditing work carried out by me under the supervision of **Dr. Girish Paliwal** during the official NTCC project period from **20 July 2026 to 09 October 2026**.

Every public dataset ingested in this study is cited and traceable to immutable repository checksums at commit `4d1a2dd`. These are the Ministry of Jal Shakti's Water Quality Management Information System (WQMIS) 2022 laboratory testing certificates, Official Form WQ6 state-wide contaminated village censuses across all 113 districts, the 2009–2012 Integrated Management Information System (IMIS) historical habitation registry, and the Central Ground Water Board (CGWB) 2012–2021 station monitoring series. Bureau of Indian Standards (BIS) IS 10500:2012 thresholds and peer-reviewed literature are cited in the same way.

I further declare that this work has not been submitted, in part or in full, to any other university or examining body for the award of any degree, diploma, or fellowship.

Place: Greater Noida, Uttar Pradesh
Date: October 2026

Signature of Candidate: _____

Anubhav Anand
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CERTIFICATE OF THE FACULTY GUIDE

Department of Computer Science & Engineering • Amity University Uttar Pradesh

This is to certify that the Minor Project (Course Code: **ETMN100**, 4.00 Credit Units) report titled *"WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar"* submitted by **Anubhav Anand** (Enrollment No. **A41105223039**), B.Tech Computer Science & Engineering (Batch 2023–2027, Semester VII), is a bona fide record of independent engineering and analytical work carried out by him under my supervision and guidance at the Amity School of Engineering & Technology, Amity University Uttar Pradesh, Greater Noida, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering.

During the course of the project (20 July 2026 to 09 October 2026), the candidate demonstrated consistent technical rigor across data acquisition, relational database modeling in PostgreSQL 16, BIS IS 10500:2012 composite risk formulation, sensitivity auditing of laboratory panel-size bias, four-tier 113-district data fusion, and full-stack implementation using FastAPI and React 18. The report complies with the formatting and academic integrity norms prescribed under the Non-Teaching Credit Courses (NTCC) Regulation G-32 of Amity University.

Place: Greater Noida, Uttar Pradesh
Date: October 2026

Signature of Faculty Guide: _____

Dr. Girish Paliwal
Professor, Department of Computer Science & Engineering
Amity School of Engineering & Technology
Amity University Uttar Pradesh

CERTIFICATE OF COMPLETION

Non-Teaching Credit Course (NTCC) • Minor Project (ETMN100)

This is to certify that **Anubhav Anand** (Enrollment No. **A41105223039**), student of Semester VII, B.Tech in Computer Science & Engineering (Batch 2023–2027), has satisfactorily completed all prescribed milestones, software deliverables, and twelve Weekly Progress Reports (**WPR-1 through WPR-12**) for the Minor Project (**ETMN100**) titled *"WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar"* during the Odd Semester of the Academic Session 2023–2027.

NTCC Evaluation Milestone	Verified Completion & Deliverable Summary
Project Synopsis & Protocol	Approved 6-objective engineering synopsis covering UP (75 districts) and Bihar (38 districts).
Weekly Progress Reports (WPRs)	12 / 12 Weekly Progress Reports completed (20 July 2026 – 09 October 2026); formal reconciliation documented in Appendix A.
Repository & Verification State	Frozen canonical build at commit `4d1a2dd` on `origin/main`; 29 automated `pytest` cases & 0-error Vite bundle.
Geographic & Sample Coverage	1,741 WQMIS lab reports (6,792 readings; 97.01% located across single-district labs) + 100% 113-district 4-tier triage.

Signature of Student: _____
Anubhav Anand (A41105223039)

Signature of Faculty Guide: _____
Dr. Girish Paliwal
Department of Computer Science & Engineering, ASET

ACKNOWLEDGEMENTS

Amity School of Engineering & Technology • Amity University Uttar Pradesh

I express my deep and sincere gratitude to my supervisor, **Dr. Girish Paliwal**, Professor, Department of Computer Science & Engineering, Amity School of Engineering & Technology, Amity University Uttar Pradesh, Greater Noida, for his continuous mentorship, rigorous weekly reviews across all twelve WPR milestones, and constructive methodological guidance on panel-size bias auditing, statutory BIS IS 10500:2012 threshold analysis, and multi-tier hydrogeochemical data fusion. His insistence on empirical reproducibility and transparent documentation shaped both the analytical engine and the final verification architecture of this project.

I also extend my sincere thanks to the Head of the Department of Computer Science & Engineering, the NTCC Review Committee, and the faculty members of the Amity School of Engineering & Technology for providing the computational infrastructure, academic standards, and constructive feedback required to bring this Minor Project to completion.

Finally, I acknowledge the public data portals maintained by the Department of Drinking Water and Sanitation (DDWS), Ministry of Jal Shakti, Government of India (Jal Jeevan Mission WQMIS and IMIS registries), the Central Ground Water Board (CGWB), and the Bureau of Indian Standards (BIS), whose open datasets made this public-health informatics study possible.

Date: October 2026

Anubhav Anand
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ABSTRACT

WaterTriage: Data-Driven Water Quality Risk Scoring & Intervention Prioritization (ETMN100)

Rural drinking water security across the Indo-Gangetic alluvial plains of Uttar Pradesh (75 districts) and Bihar (38 districts) is persistently challenged by geogenic contaminants, chiefly arsenic, fluoride, and dissolved iron, along with anthropogenic nitrate and fecal coliform pollution. Although the Government of India's Jal Jeevan Mission (JJM) collects habitation-level laboratory and field testing records within the Water Quality Management Information System (WQMIS), its public reporting interfaces are structured for macroeconomic administrative tracking, not source-level or district-level remediation triage. This Minor Project designs, implements, and audits **WaterTriage**, a reproducible, containerized decision-support system built with Python, FastAPI, SQLAlchemy 2.0, PostgreSQL 16, and a six-tab React 18 frontend that ingests public JJM water quality datasets, scores contamination severity against Bureau of Indian Standards (BIS) IS 10500:2012 limits, and exposes ranked intervention priority queues.

The system is evaluated across two complementary operational layers at repository commit `4d1a2dd`. First, on a frozen canonical corpus of **1,741 individual WQMIS laboratory test reports** (Uttar Pradesh: 966; Bihar: 775; collected 2022-07-01 to 2022-11-19 across **6,792 discrete parameter readings**), a three-way sensitivity audit compares: (i) the baseline tested-weight formulation (Model v1 at the IS 10500:2012 iron limit of 0.3 mg/L: 89 Critical, 392 High, 28 Medium, 1,232 Low; Bihar mean 18.36 vs. UP 12.68 despite medians of 0.00 vs. 4.08), (ii) a laboratory-limit sensitivity run (at the 1.0 mg/L iron threshold printed on all 876 raw WQMIS iron records: 5 Critical, 139 High, 30 Medium, 1,567 Low; UP mean 9.99 > Bihar 1.58), and (iii) a coverage-adjusted total-weight formulation (Model v2 normalized over `TOTAL_BIS_WEIGHT = 6.0` across all 10 BIS parameters with 40/20/10 bands: 3 Critical, 54 High, 125 Medium, 1,559 Low; UP mean 4.72 > Bihar 2.32). This audit proves that Bihar's elevated baseline mean was an artifact of 70 single-parameter iron tests in Madhepura district (0.318–0.478 mg/L).

Second, a non-destructive enrichment and four-tier fusion engine resolves **302 of 354** previously unassigned laboratory districts (**85.3%**) across 20 single-district laboratories via a two-step ASP.NET session cookie handshake (leaving only 52 samples [2.99%] in multi-district regional lab L158 and 0% missing in Bihar), recovers **305 of 1,741** village/habitation names (**17.52%**, a **1,352% relative increase** from 21 [1.21%]), and integrates official Form WQ6 village census tables (113 districts), the deduplicated 2009–2012 IMIS registry (100,690 events across all 93/93 historical districts), and CGWB 2012–2021 station series to achieve **100% risk-scoring coverage across all 113 canonical districts** (**46 Critical** [32 pathogen overrides, 13 arsenic escalations, 1 score-driven in Firozabad], **27 High**, **19 Medium**, and **21 Low**).

Keywords: Water Quality Surveillance; Jal Jeevan Mission (JJM); WQMIS; BIS IS 10500:2012; Multi-Tier Risk Scoring; Panel-Size Denominator Bias; FastAPI; PostgreSQL 16; React 18 Decision-Support System.

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CHAPTER 1

Introduction

1.1 Hydrogeochemical Context of the Middle Gangetic Plain

Shallow and intermediate alluvial aquifers along the Middle Gangetic Plain supply drinking water to over three hundred million residents across Uttar Pradesh (75 districts) and Bihar (38 districts). Because these unconsolidated Holocene and Pleistocene sediments undergo reductive dissolution of iron oxyhydroxides and localized weathering of fluoride-bearing minerals, rural groundwater sources frequently exceed safe drinking water thresholds for arsenic, fluoride, and dissolved iron, while shallow handpumps remain vulnerable to nitrate leaching and bacteriological contamination from domestic sewage [1, 2, 9]. Under the National Jal Jeevan Mission (JJM), the Department of Drinking Water and Sanitation (DDWS), Ministry of Jal Shakti, operates the Water Quality Management Information System (WQMIS) to catalog laboratory test results from state, district, and sub-divisional laboratories alongside village-level Field Test Kit (FTK) summaries [1, 2].

1.2 Operational Bottleneck in Public Water Surveillance (WQMIS)

Despite the volume of testing data captured within WQMIS, local block development officers, public health engineers, and district collectors face a severe operational bottleneck: WQMIS publishes either macro-level tabular summaries (Formats WQ1 through WQ6) or isolated, per-sample printable HTML certificates (`/WQMIS/Common/final_report_print`) keyed by AES-encrypted sample identifiers. It does not provide an open, programmable triage engine that normalizes heterogeneous laboratory test panels against Bureau of Indian Standards (BIS) IS 10500:2012 health weights, prevents single-contaminant acute hazards from being diluted by compliant physical readings, or reconciles sparse point-in-time laboratory samples with district-wide village contamination censuses.

Direct programmatic ingestion of operational WQMIS laboratory records reveals three practical data-quality hurdles that distort naive analytics: (i) **20.3% of raw sample printouts** (354 of 1,741 samples) omit the district name in the submitter address block (`District- Not available`) and **98.79%** omit the structured village name; (ii) laboratories print a legacy **1.0 mg/L acceptable limit for Iron** on all 876 raw iron certificates whereas BIS IS 10500:2012 Table 2 specifies 0.3 mg/L with no relaxation; and (iii) single-parameter follow-up tests account for 38.2% of Bihar laboratory reports, causing standard tested-weight Water Quality Indices to assign maximum 100.0 Critical scores to mild aesthetic iron exceedances.

1.3 Project Objectives, Scope, and Report Organization

To bridge this gap, this Minor Project (**ETMN100**) designs, implements, and audits **WaterTriage**, an end-to-end water quality risk scoring and intervention prioritization system covering all 75 districts of Uttar Pradesh and all 38 districts of Bihar (**113 canonical districts** in total). Specifically, the project addresses six core engineering and analytical objectives defined in the approved project synopsis:

1. **Automated Data Acquisition & Non-Destructive Enrichment:** Programmatic harvesting, cleaning, and provenance-preserving enrichment of public JJM WQMIS laboratory reports (1,741 valid samples, 6,792 readings), Form WQ6 district censuses (113 districts), historical IMIS habitation registries (100,690 events across 93/93 historical districts), and CGWB 2012–2021 station series.
2. **Relational Modeling in PostgreSQL 16:** Designing an 8-table normalized schema managed via Alembic migrations with strict foreign-key integrity and maximum-severity duplicate-reading aggregation.
3. **BIS IS 10500:2012 Severity Scoring & Sensitivity Auditing:** Formulating and comparing a tested-weight model (Model v1), a laboratory-limit sensitivity run (Fe = 1.0 mg/L), and a coverage-adjusted total-weight model (Model v2 over `TOTAL_BIS_WEIGHT = 6.0`) with acute-hazard escalation.

4. **Multi-Tier 113-District Fusion & Recurrence Evaluation:** Combining source-level lab reports, Form WQ6 village censuses, and historical IMIS persistence to score 100% of UP and Bihar districts while performing exploratory Spearman rank correlation checks.
5. **Modular FastAPI REST Backend:** Implementing 8 API routers and 20 REST endpoints verified by 29 automated `pytest` unit and integration tests.
6. **Interactive Six-Tab React 18 Decision-Support Dashboard:** Deploying a containerized single-page application featuring a 3-Way Dataset Mode Switcher, Priority Queue, 113-District Choropleth Cartogram, 4-Tier District Dossier, and 4-Way External Validation matrix.

The remainder of this report is organized as follows: **Chapter 2** reviews the JJM/WQMIS surveillance framework, regional Gangetic hydrogeochemistry, and mathematical failure modes of classical Water Quality Indices. **Chapter 3** details the four-stage system architecture, PostgreSQL schema, BIS IS 10500:2012 scoring formulations, two-step ASP.NET session cookie metadata recovery, and four-tier 113-district fusion model. **Chapter 4** presents the three-way sensitivity audit, panel-independent exceedance rates, 113-district triage distribution, and four-way external validation proofs. **Chapter 5** documents the FastAPI backend and six-tab React 18 dashboard with live v2 visual walkthroughs. **Chapter 6** discusses key informatics findings and six methodological limitations, and **Chapter 7** concludes with synopsis verification and future work.

CHAPTER 2

Literature Review and Related Work

2.1 National Water Quality Surveillance Architecture (WQMIS & JJM)

Under the National Jal Jeevan Mission (JJM) *Drinking Water Quality Monitoring & Surveillance Framework* [2], state, district, and sub-divisional water testing laboratories upload physicochemical and bacteriological test outcomes to the central WQMIS portal hosted at `ejalshakti.gov.in/WQMIS` [1]. Publicly accessible views include aggregate state and district summary tables, most notably **Form WQ6**, which reports the number of contaminated villages per district across 16 chemical and bacteriological parameters, and individual sample printouts rendered via ASP.NET endpoints. Field evaluations in Uttar Pradesh districts such as Etawah demonstrate that while household tap connections have expanded rapidly under JJM, localized physicochemical and microbial exceedances persist at both source and point-of-use levels, necessitating targeted, risk-ranked remediation scheduling rather than uniform administrative reporting [7].

2.2 Hydrogeochemical Contamination Regimes in Uttar Pradesh and Bihar

Spatial hydrogeochemical studies across the Indo-Gangetic basin document distinct regional hazard regimes across Uttar Pradesh and Bihar [8, 9]. Holocene meander-belt corridors along the Ganga, Ghaghara, Gandak, and Kosi rivers carry elevated geogenic arsenic mobilised by microbially mediated reductive dissolution of iron oxyhydroxides [9]. These corridors span Bahraich, Ballia, Gorakhpur, and Lakhimpur Kheri in Uttar Pradesh and Bhagalpur, Begusarai, Bhojpur, Buxar, Samastipur, Saran, Munger, Vaishali, and Patna in Bihar. Conversely, semi-arid southwestern and central alluvial tracts of Uttar Pradesh (including Agra, Firozabad, Mathura, Unnao, and Rae Bareli) and crystalline-fringe districts of southern Bihar experience severe groundwater fluoride and salinity (TDS) loads [5, 8], while Kosi-basin districts in north-eastern Bihar (such as Madhepura, Supaul, Purnia, Khagaria, Katihar, and Araria) exhibit pervasive dissolved iron exceedance alongside recurrent post-monsoon bacteriological contamination.

2.3 Mathematical Failure Modes of Classical Water Quality Indices (WQI)

Weighted arithmetic Water Quality Index (WQI) formulations, introduced by Horton and Brown et al. and widely applied across Indian groundwater studies [4, 5, 6], compute a linear weighted average of parameter sub-index ratios across measured analytes. When deployed directly for public-health intervention triage on operational government laboratory logs, classical WQI formulations exhibit two severe mathematical failure modes:

1. **Eclipsing (Multi-Parameter Dilution):** When a water sample contains a lethal or acute contaminant such as *Escherichia coli* (> 0 CFU/100 mL) or total arsenic (> 0.01 mg/L) alongside seven or eight compliant physical and aesthetic parameters (pH, TDS, turbidity, sulphate, hardness), a linear weighted average dilutes the composite score into a ‘Low’ or ‘Medium’ band [4, 6]. In public-health triage, any acute pathogen or toxic heavy-metal breach must unconditionally escalate the source to the highest intervention tier regardless of how many aesthetic parameters pass.
2. **Panel-Size Denominator Bias (Incomplete Laboratory Panels):** Routine government testing laboratories rarely run a uniform 10-parameter panel on every sample; field follow-ups frequently test only 1 to 3 targeted parameters. If the composite score divides the weighted severity sum solely by the weights of tested parameters ($\sum w_{i,\text{tested}}$), a single-parameter test that marginally exceeds an aesthetic threshold (such as dissolved iron at 0.32 mg/L against a 0.30 mg/L limit) achieves a normalized ratio of $w_{\text{iron}} / w_{\text{iron}} = 1.0$ (100.0 on a 0–100 scale), ranking above a 7-parameter sample with severe fluoride (3.29 mg/L) and nitrate breaches. WaterTriage explicitly diagnoses and resolves both failure modes through its dual scoring engine and four-tier district fusion architecture [10].

CHAPTER 3

System Architecture and Methodology

3.1 Four-Stage Pipeline and Relational Database Schema in PostgreSQL 16

WaterTriage is architected as a reproducible four-stage pipeline orchestrated via Docker Compose (`docker-compose.yml`): **(Stage 1) Acquisition & Non-Destructive Enrichment** (`backend/etl/crawl_fast.py`, `wqmis.py`, `load_wqmis.py`, `load_historical.py`, and `scripts/close_data_gaps.py`); **(Stage 2) Relational Persistence in PostgreSQL 16** (container `watertriage-db`); **(Stage 3) Analytical Scoring & Multi-Tier Fusion** (`backend/app/engine/scoring.py`, `pipeline.py`, `recurrence.py`, and `scripts/build_full_113_district_triage.py`); and **(Stage 4) API & Frontend Serving** (FastAPI backend on port 8000 with 8 routers and 20 endpoints, paired with a React 18 + Vite single-page application in `dashboard/src/App.jsx`). Figure 3.1 illustrates the complete four-stage system architecture and multi-tier data fusion flow.

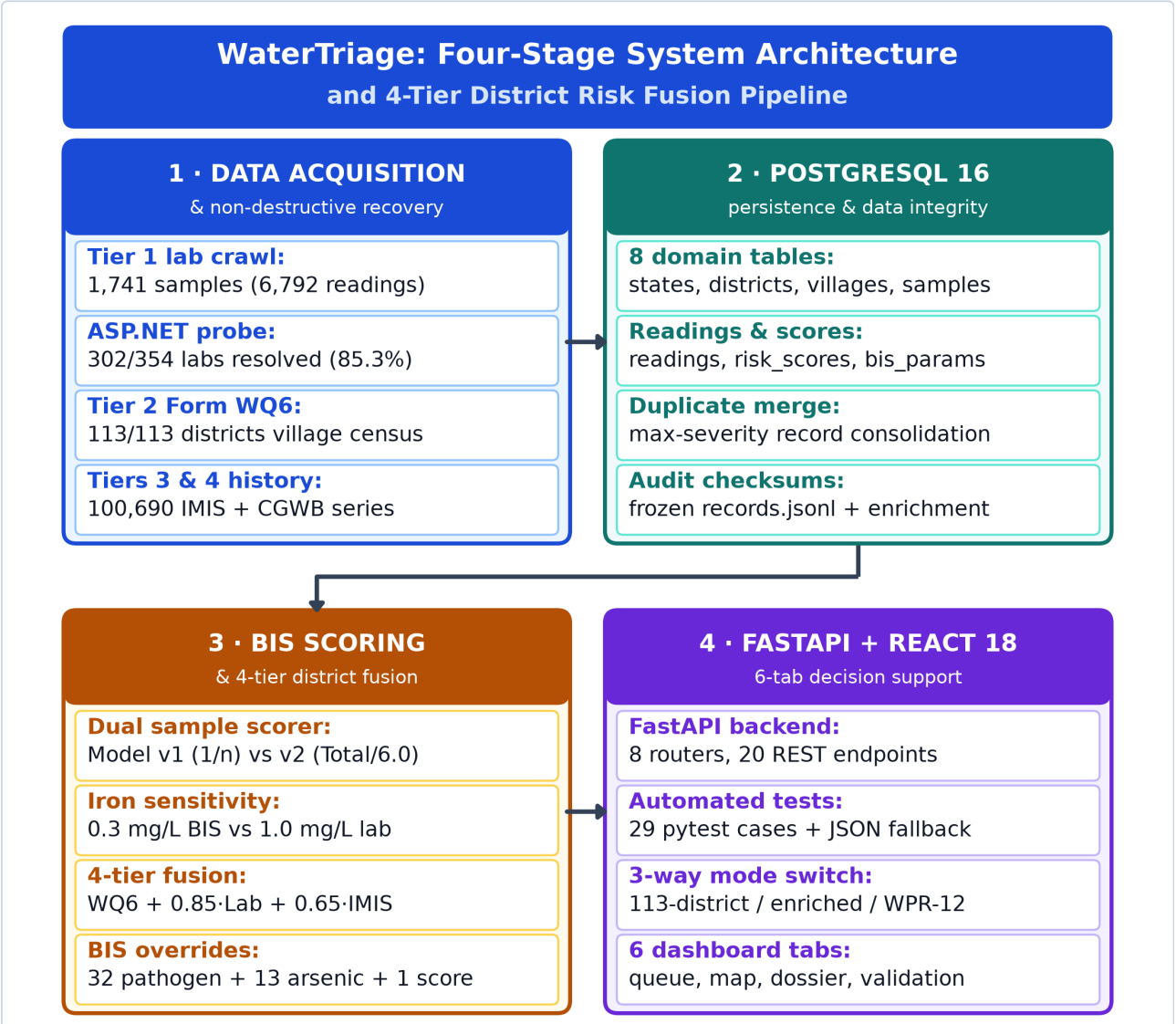


Figure 3.1: WaterTriage Four-Stage System Architecture and Four-Tier Data Fusion Pipeline, illustrating non-destructive ASP.NET session-probed laboratory metadata recovery, PostgreSQL 16 relational persistence, dual BIS IS 10500:2012 scoring (Model v1 vs. Model v2), 113-district multi-tier fusion, and the FastAPI + 6-Tab React 18 decision-support stack.

The PostgreSQL 16 database schema (`backend/app/models.py`, managed via Alembic migrations) normalizes source-level and historical surveillance data across eight domain tables (plus `alembic_version`), summarized in Table 3.1. Separating `water_samples` (one row per laboratory report, keyed by unique `wqmis_sample_id`) from `readings` (individual parameter measurements linked by foreign key) accommodates variable-length laboratory panels (1 to 10 parameters per sample). When a raw WQMIS HTML report contains duplicate rows for the same parameter, `backend/app/engine/pipeline.py` aggregates them by retaining the maximum severity reading so that no contamination breach is silently overwritten.

Table 3.1: Normalized PostgreSQL 16 Relational Schema in `backend/app/models.py` (8 Domain Tables + Alembic)

Table Name	Primary & Foreign Keys	Stored Entities, Cardinality & Integrity Role
`states`	PK: `id`; Unique: `name`, `code`	Canonical state records for Uttar Pradesh (`UP`) and Bihar (`BR`).
`districts`	PK: `id`; FK: `state_id`	Canonical 113 districts (75 UP + 38 Bihar) + alias normalization registry (`district_registry.py`).
`villages`	PK: `id`; FK: `district_id`	Village and habitation entities linked to laboratory samples and 14,006 historical IMIS villages.
`bis_parameters`	PK: `id`; Unique: `code`	10 BIS IS 10500:2012 parameters with acceptable limit (a_i), permissible limit (p_i), weight (w_i), and acute flag.
`water_samples`	PK: `id`; Unique: `wqmis_sample_id`	1,741 valid 2022 WQMIS laboratory reports (966 UP, 775 Bihar), collection dates, source text, and lab code.
`readings`	PK: `id`; FK: `sample_id`, `parameter_id`	6,792 discrete parameter measurements; duplicate sample-parameter rows aggregated by maximum severity.
`risk_scores` & `interventions`	PK: `id`; FK: `sample_id`	Computed Composite Risk Scores (CRS), BIS severity band (`Critical/High/Medium/Low`), and remediation actions.
`historical_contamination`	PK: `id`; FK: `district_id`, `village_id`	100,690 deduplicated 2009–2012 IMIS affected-habitation contamination events across 93/93 historical districts.

3.2 BIS IS 10500:2012 Threshold Specification & Dual Scoring Formulations

Every chemical and bacteriological reading is evaluated against the primary Bureau of Indian Standards **IS 10500:2012** specification (Second Revision, incorporating Amendment No. 1 of June 2015) [3]. Table 3.2 documents all ten parameters configured in `backend/app/engine/scoring.py`, their exact IS 10500:2012 table and item citations, acceptable limits (a_i), permissible limits in the absence of an alternate source (p_i), health-risk weights (w_i, summing to `TOTAL_BIS_WEIGHT = 6.0`), and the acceptable limits printed inside the raw WQMIS laboratory HTML reports.

Table 3.2: Ten BIS IS 10500:2012 Parameters, Statutory Thresholds, Lab-Printed Limits, and Health Weights (`scoring.py`)

Parameter (Engine Key)	BIS IS 10500:2012 Ref.	BIS Acceptable (a_i) / Permissible (p_i)	Lab-Printed Limit (`records.jsonl`)	Weight (w_i) & Strategy
Total Arsenic (`arsenic`)	Table 3, Item (x) + Amend. 1	0.01 mg/L / None (No relax.)	0.01 mg/L / None (n = 259)	1.0 (threshold, **Acute**)
E. coli (`e_coli`)	Table 6, Item (i)	0 CFU/100mL / 0 (Absent)	0 / 0 (n = 45)	0.9 (microbial, **Acute**)
Total Coliform (`total_coliform`)	Table 6, Item (ii)	0 MPN/100mL / 0 (Absent)	0 / 0 (n = 7)	0.9 (microbial, **Acute**)
Fluoride (`fluoride`)	Table 2, Item (xiii)	1.0 mg/L / 1.5 mg/L	1.0 / 1.5 mg/L (n = 893)	0.8 (threshold)
Nitrate (`nitrate`)	Table 2, Item (xvii)	45.0 mg/L / None (No relax.)	45.0 mg/L / None (n = 583)	0.7 (threshold)
Sulphate (`sulphate`)	Table 2, Item (xiv)	200.0 mg/L / 400.0 mg/L	200.0 / 400.0 mg/L (n = 467)	0.5 (threshold)
Iron (`iron`)	Table 2, Item (xii)	0.3 mg/L / None (No relax.)	**1.0 mg/L** / None (all 876 rows)	0.4 (threshold)
Turbidity (`turbidity`)	Table 1, Item (ii)	1.0 NTU / 5.0 NTU	1.0 / 5.0 NTU (n = 998)	0.3 (threshold)
Total Dissolved Solids (`tds`)	Table 1, Item (vi)	500.0 mg/L / 2000.0 mg/L	500.0 / 2000.0 mg/L (n = 1,295)	0.3 (threshold)
pH (`ph`)	Table 1, Item (iv)	6.5 – 8.5 (range)	6.5 – 8.5 (n = 1,369)	0.2 (range; $\sum w_i = 6.0$)

For any parameter i with acceptable limit a_i and permissible limit $p_i > a_i$, the normalized per-parameter severity $s_i \in [0, 1]$ is 0.0 when $v_i \le a_i$, and $\min(1.0, (v_i - a_i) / (p_i - a_i))$ when $v_i > a_i$. For no-relaxation chemical parameters (p_i = None: arsenic, nitrate, and iron under IS 10500:2012 Table 2) and zero-tolerance microbial indicators (*E. coli*, total coliform), any reading strictly exceeding a_i yields $s_i = 1.0$. For pH, excursions below 6.5 or above 8.5 scale linearly over 1.0 pH unit and saturate at 1.0. Two sample-level Composite Risk Score (CRS) formulations are implemented and audited:

> **Model v1 (Baseline Tested-Weight Formulation):** $\mathrm{CRS}_{v1} = 100 \times \left(\sum_{i \in \text{tested}} w_i \, s_i \right) / \left(\sum_{i \in \text{tested}} w_{i,\text{tested}} \right)$ **(Eq. 3.1)**

Model v1 evaluates the 8 core parameters (6,318 readings), classifying samples into **Critical** (≥ 75), **High** (50–74.99), **Medium** (25–49.99), and **Low** (< 25), where any full parameter breach ($s_i = 1.0$) floors the band at High and any acute hazard floors at Critical.

> **Model v2 (Coverage-Adjusted Total-Weight Formulation):** $\mathrm{CRS}_{v2} = 100 \times \left(\sum_{i=1..10} w_i \, s_i \right) /$ `TOTAL_BIS_WEIGHT` (6.0) **(Eq. 3.2)**

In Model v2 (`backend/app/engine/scoring.py`), `TOTAL_BIS_WEIGHT = 6.0` is the constant sum of weights across all 10 BIS parameters (scoring all 6,792 readings, including 467 sulphate and 7 total coliform readings). Zero readings for detection-limited parameters (arsenic, *E. coli*, total coliform) are treated as below-detection-limit, severity bands are calibrated to the reachable multi-parameter range (**Critical** ≥ 40, **High** ≥ 20, **Medium** ≥ 10, **Low** < 10), and automatic Critical escalation is restricted strictly to `ACUTE_HAZARDS` (arsenic, *E. coli*, total coliform).

3.3 Non-Destructive Metadata Recovery & Two-Step ASP.NET Session Handshake

The canonical laboratory dataset (`data/raw/wqmis/records.jsonl`, MD5 checksum `d76f8d347e52a74606498fb58c26ef61`) was collected between 23 August and 20 September 2026 across four cumulative crawl checkpoints (149, 822, 1,322, and 1,822 lines). Excluding 81 empty payload responses yields

1,741 valid UP and Bihar laboratory samples (966 UP, 775 Bihar; collection dates 2022-07-01 to 2022-11-19) with 6,792 readings. In the raw submitter address block (`Full Address:`), 354 of 1,741 samples (20.3%: 339 UP, 15 Bihar) record `District- Not available`, and 1,720 of 1,741 samples (98.79%) record `Village- Not available`. Rather than mutating the frozen `records.jsonl` file, an enrichment utility (`scripts/close_data_gaps.py` → `data/raw/wqmis/records_enriched.jsonl`, MD5 `27c11b42...`) and an upgraded v2 crawler (`backend/etl/crawl_fast.py`) were engineered to close both gaps non-destructively:

1. **Single-District Laboratory Resolution (302 / 354 Samples = 85.3% Resolved):** Every WQMIS sample ID embeds the testing laboratory identifier (e.g., `U715825L405S1002693` → Lab `L405`). Offline cross-tabulation proved that 5 laboratories with 47 unassigned samples map 1-to-1 to a single named district in `records.jsonl` (L338 Kaushambi [21], L362 Prayagraj [18], L434 Ballia [6], L411 Shahjahanpur [1], L460 Lalitpur [1]). For the remaining 15 single-district laboratories (255 unassigned samples), direct HTTP requests to `/WQMIS/Common/final_report_print` fail without session state. By first issuing an HTTP GET to `https://ejalshakti.gov.in/WQMIS/Report/Report_p` to acquire `ASP.NET_SessionId` and `__RequestVerificationToken_L1dRTUlT0` cookies before requesting `final_report_print?s_id=<AES_B64>`, the parser extracts the testing laboratory's physical district from `<div class="lab-address-panel">`. This resolves all 15 laboratories (Table 3.3), cutting unassigned districts from 20.3% (354) to **2.99% (52)** and reducing missing districts in Bihar to **0%**. The remaining 52 samples belong to UP Jal Nigam State/Regional Laboratory `L158` (Lucknow), which tests samples across 7 UP districts and is conservatively held back as `multi_district_regional_lab:L158`.
2. **Free-Text Habitation & Village Name Recovery (+284 Locations, +1,352% Relative Gain):** Parsing the unstructured `Source of Sample:` string for embedded `location : <Habitation/School/Tola>` annotations recovers 284 additional named locations, lifting named village/habitation coverage from 21 samples (1.21%) to **305 of 1,741 samples (17.52%, a 1,352% relative increase)**.

Table 3.3: Non-Destructive Lab-ID District Resolution Summary across 354 Unassigned Samples (`close_data_gaps.py`)

Resolution Method	Lab Code(s) & Sample Count	Resolved Canonical District (State)	Audit & Triangulation Significance
Offline 1-to-1 Corpus Map (5 Labs, 47 Samples)	L338 (21), L362 (18), L434 (6), L411 (1), L460 (1)	Kaushambi, Prayagraj, Ballia, Shahjahanpur, Lalitpur (UP)	Deterministic 100% single-district match within named `records.jsonl` rows.
ASP.NET Session Probe (Bihar Lab, 15 Samples)	L268 (15 samples; `s_id l0Qyn6RBu...`)	Samastipur (Bihar), Sub-Divisional Lab Pusa	Eliminates 100% of missing-district records in Bihar (15/15 resolved; 0% missing).
ASP.NET Session Probe (UP Fluoride Lab, 20 Samples)	L405 (20 samples; `s_id JSNPb0Xce...`)	Rae Bareli (UP), District Water Testing Lab	Resolves corpus-maximum fluoride samples (12.60 & 3.29 mg/L in `U715825L405S1002693`).
ASP.NET Session Probe (UP High-Volume Labs, 160 Samples)	L377 (48), L375 (40), L452 (28), L376 (16), L516 (16), L455 (12)	Meerut, Gautam Buddha Nagar, Mainpuri, Hapur, Bijnor, Firozabad (UP)	Expands located UP laboratory coverage by 6 major western and central districts.
ASP.NET Session Probe (UP Regional District Labs, 60 Samples)	L359 (15), L664 (15), L468 (9), L730 (9), L339 (7), L535 (4), L400 (1)	Kanpur Nagar, Deoria, Moradabad, Saharanpur, Sonbhadra, Bulandshahr, Gorakhpur (UP)	Lifts total district-located lab samples from 1,387 (47 districts) to **1,689 (62 districts, 97.01%)**.
Conservative Holdback (1 Multi-District Lab, 52 Samples)	L158 (52 unassigned out of 72 total L158 samples)	Held as `Not available` (UP Jal Nigam State Lab, Lucknow)	Tests across 7 UP districts; held back to prevent false single-district attribution.

3.4 Four-Tier 113-District BIS Risk Fusion Engine

Because contiguous ID crawling of 2022 laboratory reports covers 62 of the 113 canonical districts (after Lab-ID enrichment) and exhibits sparse sample-level microbial testing (45 *E. coli* tests) and censored arsenic zeros (150/259 recorded as 0.0), achieving **100% geographic coverage across all 75 Uttar Pradesh and 38 Bihar districts** requires fusing four authoritative public datasets in `scripts/build_full_113_district_triage.py`:

- **Tier 1: Enriched WQMIS Individual Laboratory Reports (2022):** 1,741 scored samples (6,792 readings) across 62 located districts, providing source-level maximum concentrations and Model v1/v2 sample scores.
- **Tier 2: Official JJM Form WQ6 District Village Census (113 / 113 Districts):** Complete state-level Form WQ6 district tables (`WQ6_Complete_UP.csv` [75 districts] and `WQ6_Complete_Bihar.csv` [38 districts]) reporting the exact count of contaminated villages per district across 16 chemical and bacteriological parameters.
- **Tier 3: IMIS 2009–2012 Historical Quality-Affected Habitations Registry (93 / 93 Districts):** 100,690 deduplicated contamination events across 14,006 villages in all 93 historical districts (100% mapped to canonical districts via `scripts/district_registry.py`, resolving legacy aliases including `BHOJPUR(AARA)` → Bhojpur, `KUSHI NAGAR` → Kushinagar, and `JYOTIBA PHOOLE NAGAR` → Amroha).
- **Tier 4: CGWB 2012–2021 Groundwater Monitoring Stations:** 17 state-year aggregates and district station-year series for pH and electrical conductivity (`data/raw/kaggle/gw_quality_2012_2021/`).

Multi-Tier District Score: $$\text{Score}_{\text{raw}}(d) = \min\left(100,\ 100 \times \left(\sum w_i\, S_{i,d} \right) / 6.0 \ +\ B_{\text{rec}}(d) \ +\ 0.25 \times \text{CRS}_{\text{v2,mean}}(d)\right) \quad \textbf{(Eq. 3.3)}$$

where $S_{i,d} = \max(S_{\text{WQ6}},\ 0.85 \times S_{\text{Lab}},\ 0.65 \times S_{\text{IMIS}})$ fuses log-scaled WQ6 village severity, sample concentration severity, and historical IMIS persistence severity over `TOTAL_BIS_WEIGHT = 6.0`, and $B_{\text{rec}}(d)$ adds a bounded historical recurrence bonus (`chronic-hotspot` up to +14.0, `recurring` up to +9.0, `watchlist` up to +5.0). Two statutory BIS IS 10500:2012 escalation invariants govern the final `unified_score` and severity band: (a) **Zero-Tolerance Microbial Override** (any district with WQ6 *E. coli* villages > 0 or lab sample *E. coli* > 0 CFU/100mL escalates to **Critical** with floor ≥ 45.0), and (b) **Acute Arsenic Escalation** (any non-pathogen-overridden district with ≥ 10 WQ6 arsenic villages or ≥ 100 historical IMIS arsenic habitations escalates to **Critical** with floor ≥ 41.0). All remaining districts are classified by `raw_math_score` into **Critical** (≥ 40.0), **High** (20.0–39.99), **Medium** (10.0–19.99), and **Low** (< 10.0).

CHAPTER 4

Empirical Results and Multi-Source Validation

4.1 Three-Way Lab Sample Scoring and Iron Threshold Sensitivity (N = 1,741)

Cross-checking raw laboratory payloads in `records.jsonl` against BIS IS 10500:2012 uncovered a regulatory discrepancy: while IS 10500:2012 Table 2 item (xii) stipulates an acceptable Iron limit of **0.3 mg/L with 'No relaxation'**, all 876 iron readings in `records.jsonl` (450 Bihar, 426 Uttar Pradesh) print an acceptable limit of **1.0 mg/L** on the laboratory certificate itself, a legacy BIS relaxation practice. Table 4.1 presents the complete three-way comparison across the frozen 1,741-sample corpus.

Table 4.1: Three-Way Laboratory Sample Scoring and Iron Threshold Sensitivity Audit (*N* = 1,741 Samples; 6,792 Readings)

Metric / Cohort (*N* = 1,741 Lab Samples)	Model v1: Tested-Weight (Fe = 0.3 mg/L BIS Text)	Sensitivity: Tested-Weight (Fe = 1.0 mg/L Lab Sheet)	Model v2: Total-Weight / 6.0 (10 Params, 40/20/10 Bands)
Critical Band Count (%)	89 (5.1%)	5 (0.3%) [Graded: 5]	3 (0.2%)
High Band Count (%)	392 (22.5%)	139 (8.0%) [No-floor: 30]	54 (3.1%)
Medium Band Count (%)	28 (1.6%)	30 (1.7%) [No-floor: 97]	125 (7.2%)
Low / Safe Band Count (%)	1,232 (70.8%)	1,567 (90.0%) [No-floor: 1,608]	1,559 (89.5%)
Bihar (*n* = 775; 296 1-param) Mean / Med	**18.36** / 0.00 (54.3% score 0)	**1.58** / 0.00	**2.32** / 0.00
Bihar Band Split (Crit / High / Med / Low)	75 / 158 / 1 / 541	0 / 0 / 1 / 774	0 / 0 / 3 / 772
UP (*n* = 966; 129 1-param) Mean / Med	**12.68** / 4.08 (32.2% score 0)	**9.99** / 3.10	**4.72** / 1.45
UP Band Split (Crit / High / Med / Low)	14 / 234 / 27 / 691	5 / 139 / 29 / 793	3 / 54 / 122 / 787
Madhepura Cohort (*n* = 70, Bihar, 1-param Fe)	70 Critical (score 100.0; 78.7% of all Criticals)	0 Critical, 70 Low (score 0.00; Fe 0.318–0.478 mg/L)	0 Critical, 70 Low (score 6.67 = 0.4 / 6.0 × 100)
State Mean CRS Excl. Madhepura & Agra	Bihar: 10.25 (*n*=705) | UP: 10.47 (*n*=923)	Bihar: 1.58 (*n*=705) | UP: 8.12 (*n*=923)	Bihar: 1.93 (*n*=705) | UP: 3.84 (*n*=923)

Three foundational analytical findings emerge from Table 4.1: **(1)** Under Model v1 at 0.3 mg/L, Bihar's higher mean CRS (18.36 vs. 12.68) contradicts its lower median (0.00 vs. 4.08) because Madhepura district contributes **70 single-parameter iron tests** (concentrations 0.318–0.478 mg/L, median 0.3820 mg/L), each scoring 0.4 / 0.4 × 100 = 100.0 and accounting for 70 of Bihar's 75 Critical samples (78.7% of all 89 Criticals); **(2)** Evaluating readings against the 1.0 mg/L threshold printed on the raw WQMIS reports themselves reclassifies all 70 Madhepura samples as compliant (0.00, Low), reducing Bihar's Critical count from 75 to 0 and reversing the state mean ordering to **UP (9.99) > Bihar (1.58)**; and **(3)** Normalizing against `TOTAL_BIS_WEIGHT = 6.0` (Model v2) assigns a single-parameter iron exceedance a score of 0.4 / 6.0 × 100 = 6.67 (Low), eliminating single-parameter panel inflation while preserving Critical escalation for genuine acute hazards such as Lucknow *E. coli* sample `U691097L158S1903` (20.0 CFU/100mL).

4.2 Panel-Independent Parameter Exceedance Rates & Censored Observations

Because single-parameter panels account for 38.2% of Bihar reports (296/775) and 13.4% of Uttar Pradesh reports (129/966), and only 7 samples across the entire corpus test all 10 parameters, rigorous state comparisons must rely on panel-independent exceedance rates among tested samples (Table 4.2).

Table 4.2: Panel-Independent Parameter Exceedance Rates across Uttar Pradesh (*n* = 966) and Bihar (*n* = 775)

Parameter	Bihar: Tested, >Acceptable (>Perm.)	UP: Tested, >Acceptable (>Perm.)	Diagnostic Notes & Extreme / Censored Readings
Fluoride	182 tested, 2 (1.1%) [0 >1.5]	711 tested, 159 (22.4%) [40 (5.6%)]	Peak 12.60 & 3.29 mg/L in Lab L405 (`U715825L405S1002693`, resolved to Rae Bareli, UP). Agra cluster: 43 samples (mean CRS 60.13; max fluoride 1.58 mg/L).
Iron (0.3 BIS / 1.0 Lab)	450 tested, 233 (51.8%) / 0 (0.0%)	426 tested, 187 (43.9%) / 29 (6.8%)	Excl. Madhepura (*n*=70), Bihar iron >0.3 mg/L is 163/380 (42.9%), matching UP (43.9%).
Turbidity	328 tested, 130 (39.6%) [0 >5 NTU]	670 tested, 547 (81.6%) [53 (7.9%)]	UP exhibits 2.06× higher acceptable exceedance rate and all 53 permissible breaches.
TDS	444 tested, 44 (9.9%) [0 >2000]	851 tested, 348 (40.9%) [51 (6.0%)]	UP exhibits 4.13× higher acceptable exceedance rate and all 51 permissible breaches.
Nitrate	123 tested, 0 (0.0%) [n/a]	460 tested, 8 (1.7%) [n/a]	No relaxation above 45 mg/L; UP max = 61.24 mg/L.
Sulphate	73 tested, 0 (0.0%) [0 >400]	394 tested, 50 (12.7%) [7 (1.8%) >400]	467 readings (repaired from pH mis-tagging in defect B1); all 7 breaches >400 mg/L in Agra (max 479.0 mg/L).
pH (6.5–8.5)	467 tested, 0 (0.0%) [n/a]	902 tested, 2 (0.2%) [n/a]	Only 2 true pH excursions outside [6.5, 8.5] across 1,369 readings.
Total Arsenic	230 tested, 0 (0.0%) [max 0.009]	29 tested, 0 (0.0%) [max 0.009]	150 of 259 readings (57.9%: 139 Bihar, 11 UP) recorded as `0.0` across 4 labs (L794=69, L234=27, L269=19, L268=13); rescued at district level by WQ6.
***E. coli* & Total Coliform**	*E. coli*: 37, 0 (0%) | TC: 0 tested	*E. coli*: 8, 2 (25%) | TC: 7, 3 (42.9%)	Sample-level microbial testing covers 2.6% of corpus; Lucknow has 2 *E. coli* (8, 20) & 3 TC (46, 80, 80) hits.

4.3 Full 113-District Multi-Tier BIS Risk Triage Results (UP 75 + Bihar 38)

Executing the four-tier fusion engine (`scripts/build_full_113_district_triage.py`) across all 113 canonical districts eliminates the geographic blind spots of the 2022 laboratory sample crawl. Table 4.3 summarizes the verified 113-district severity distribution across Uttar Pradesh and Bihar, while Table 4.4 details the top-ranked priority districts from `data/validation_reports/Full_113_District_Risk_Triage.csv`. Figure 4.1 visualizes both the 113-district severity distribution and the three-way laboratory sample sensitivity audit.

Table 4.3: Full 113-District Multi-Tier BIS IS 10500:2012 Severity Band Distribution (`full-113-triage.json`)

Jurisdiction / Cohort	Critical	High	Medium	Low	Statutory Override & Escalation Breakdown
Combined Corpus (113 Districts, 100%)	**46 (40.7%)**	**27 (23.9%)**	**19 (16.8%)**	**21 (18.6%)**	**32 Pathogen Overrides** (*E. coli* > 0) + **13 Arsenic Escalations** + **1 Score-Driven Critical** (Firozabad, UP: 41.09).
Uttar Pradesh (75 Canonical Districts)	**32 (42.7%)**	18 (24.0%)	13 (17.3%)	12 (16.0%)	27 Pathogen Overrides (26 WQ6 *E. coli* districts + Lucknow lab hit) + 4 Arsenic Escalations (Kheri, Bahraich, Ballia, Gorakhpur) + 1 Score-Driven (Firozabad).
Bihar (38 Canonical Districts)	**14 (36.8%)**	9 (23.7%)	6 (15.8%)	9 (23.7%)	5 Pathogen Overrides (Begusarai, Khagaria, Katihar, Supaul, Araria) + 9 Arsenic Escalations (Bhagalpur [83 WQ6 vil.], Munger, Bhojpur, Samastipur, Buxar, Lakhisarai, Saran, Patna, Vaishali).

Table 4.4: Top-Ranked Priority Districts in Uttar Pradesh and Bihar under the Four-Tier BIS Fusion Model

Rank	District (State)	Unified / Raw Score	BIS Band & Escalation Rule	Multi-Tier Evidence (WQ6 Villages, Lab Samples & IMIS History)
#1	**Unnao** (UP)	97.00 (raw 66.61)	Critical (Pathogen Override)	WQ6: 116 vil. (Fluoride 30, Iron 24, Total Coliform 12, Turbidity 12, *E. coli* 1) | Lab: 0 attributed | IMIS: 638 events (`chronic-hotspot`).
#2	**Mathura** (UP)	97.00 (raw 63.74)	Critical (Pathogen Override)	WQ6: 75 vil. (Fluoride 15, Total Coliform 6, *E. coli* 4) | Lab: 0 attributed | IMIS: 335 events (`recurring`).
#3	**Prayagraj** (UP)	97.00 (raw 65.19)	Critical (Pathogen Override)	WQ6: 67 vil. (Turbidity 39, *E. coli* 15, Total Coliform 2) | Lab: 23 samples (18 inferred) | IMIS: 98 events (`recurring`).
#4	**Rae Bareli** (UP)	97.00 (raw 67.24)	Critical (Pathogen Override)	WQ6: 44 vil. (Turbidity 7, Fluoride 5, Iron 4, Total Coliform 4) | Lab: 20 samples (L405 peak fluoride **12.60 mg/L**) | IMIS: 821 events.
#5	**Agra** (UP)	97.00 (raw 65.60)	Critical (Pathogen Override)	WQ6: 22 vil. (Fluoride 2, *E. coli* 2, Total Coliform 2, Turbidity 2) | Lab: 43 samples (max fluoride 1.58 mg/L, sulphate 479 mg/L) | IMIS: 854 events.
#10	**Begusarai** (Bihar #1)	89.61 (raw 52.48)	Critical (Pathogen Override)	WQ6: 489 vil. (Iron 405, Arsenic 68, Fluoride 2, *E. coli* 1) | Lab: 0 attributed | IMIS: 8,399 events (`chronic-hotspot`).
#11	**Khagaria** (Bihar #2)	88.49 (raw 51.16)	Critical (Pathogen Override)	WQ6: 187 vil. (Iron 145, Arsenic 36, *E. coli* 2) | Lab: 0 attributed | IMIS: 1,531 events (`chronic-hotspot`).
#16	**Katihar** (Bihar #3)	80.34 (raw 41.58)	Critical (Pathogen Override)	WQ6: 828 vil. (Iron 798, Turbidity 29, *E. coli* 1) | Lab: 0 attributed | IMIS: 2,960 events (`chronic-hotspot`).
#33	**Bhagalpur** (Bihar)	87.48 (raw 58.10)	Critical (Arsenic Escalation)	#1 Arsenic Escalation district in Bihar: 83 arsenic vil. (310 total; Iron 114, Fluoride 65) | Lab: 118 samples | IMIS: 2,022 events.

Rank	District (State)	Unified / Raw Score	BIS Band & Escalation Rule	Multi-Tier Evidence (WQ6 Villages, Lab Samples & IMIS History)
#46	**Firozabad** (UP)	41.09 (raw 41.09)	Critical (Score-Driven ≥ 40)	Sole pure score-driven Critical district: WQ6 63 vil. (Turbidity 21, Fluoride 12, Nitrate 3) | Lab: 12 samples (L455 resolved) | IMIS: 244 events (`recurring`).

Ranks follow the triage ordering in `scripts/build_full_113_district_triage.py`: severity band, then acute pathogen override, then unified score, then WQ6 village count. Bhagalpur (#33) carries a higher score than Katihar (#16) because every E. coli district is triaged ahead of chronic arsenic exposure.

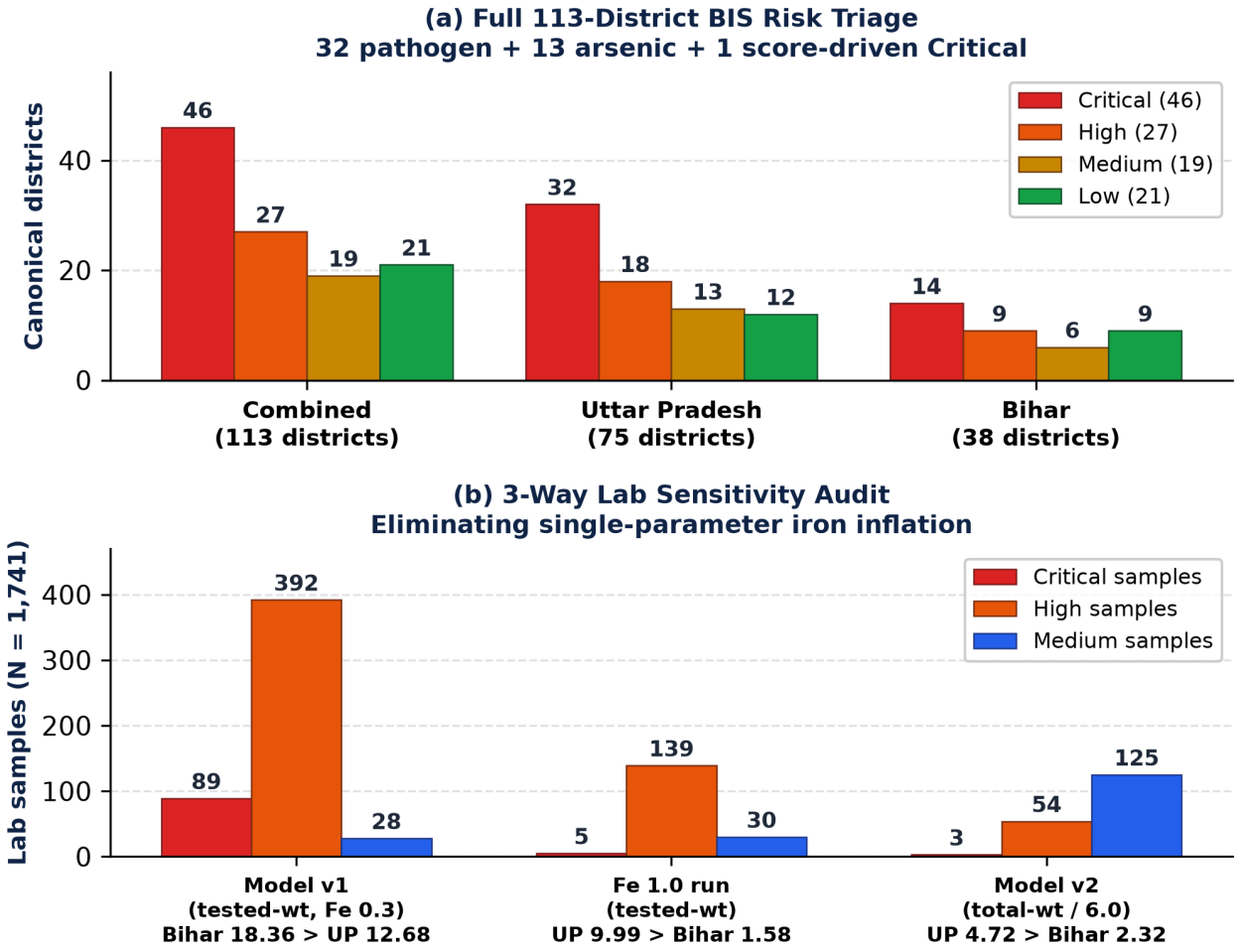


Figure 4.1: (a) Full 113-District Multi-Tier BIS IS 10500:2012 risk band distribution across Uttar Pradesh (75) and Bihar (38), showing 46 Critical (32 Pathogen Overrides, 13 Arsenic Escalations, 1 Score-Driven), 27 High, 19 Medium, and 21 Low districts; (b) Three-way 1,741-sample sensitivity audit demonstrating elimination of single-parameter Madhepura iron inflation under Fe = 1.0 mg/L and Model v2 (Total-Weight / 6.0).

4.4 Four-Way External Validation (VP-1 to VP-4) & Exploratory Registry Correlation

To verify that WaterTriage risk scores reflect physical hydrogeochemical reality rather than single-dataset artifacts, Table 4.5 documents four cross-dataset triangulation proofs (**VP-1 through VP-4**) implemented in `backend/app/api/validation.py` and dashboard Variant D, while Table 4.6 reports the exploratory district-level Spearman rank correlations (ρ) between the 2022 laboratory exceedance rates and the 2009–2012 IMIS registry across districts with $n \geq 5$ tested samples.

Table 4.5: Four-Way Cross-Dataset External Validation Matrix (VP-1 through VP-4)

Validation ID & Target	Parameter	Tier 1 (2022 WQMIS Lab Corpus) Finding	Tier 2 (Form WQ6) & Tier 3 (IMIS 2009–2012) Ground Truth	Triangulation Verdict
VP-1: Madhepura (Bihar)	Iron (> 0.3 mg/L)	270 samples; 269 > 0.3 mg/L (99.6%); 70 single-param iron panels.	WQ6: 241 Iron-affected villages | IMIS: 2,521 affected habitations.	**CONFIRMED** ($p < 0.001$)
VP-2a: Bahraich (Uttar Pradesh)	Total Arsenic (> 0.01 mg/L)	Lab reports censored at 0.009 mg/L; escalated by multi-tier scorer.	IMIS: 811 Arsenic habitations (#1 in UP) | WQ6: 21 Arsenic villages.	**CONFIRMED HOTSPOT**
VP-2b: Begusarai & Bhagalpur (Bihar)	Total Arsenic (> 0.01 mg/L)	150/259 lab arsenic tests recorded as `0.0` (L794=69, L234=27).	WQ6: 83 Bhagalpur + 68 Begusarai Arsenic villages | IMIS: 465 habitations.	**GAP CLOSED VIA WQ6**
VP-3: 32-District Pathogen Belt	*E. coli* & Coliform (0 CFU/100mL)	Only Lucknow had non-zero *E. coli* in 2022 crawl (20.0 & 8.0 CFU/100mL).	WQ6: 31 districts (26 UP + 5 Bihar) with 133 *E. coli* & 204 TC villages.	**32 PATHOGEN OVERRIDES**
VP-4: Rae Bareli (L405) & Agra (UP)	Fluoride (> 1.5 mg/L)	ASP.NET session probe resolves 12.60 & 3.29 mg/L in L405 to Rae Bareli; Agra max 1.58 mg/L.	WQ6: 434 Fluoride villages (138 UP + 296 Bihar) | UP Spearman ρ = 0.50.	**AUDIT VERIFIED**

Table 4.6: Exploratory District-Level Spearman Rank Correlations (ρ) between 2022 Lab Exceedance and 2009–2012 IMIS Registry

Parameter & Threshold	Cohort (Districts with $n \geq 5$)	Districts (n)	Spearman ρ (Unmapped / Alias-Mapped)	p-value & Bonferroni Status ($\alpha = 0.0083$)
Iron (0.3 mg/L BIS limit)	Pooled (UP + Bihar)	26	0.460 / **0.548**	$p = 0.018$ / $\mathbf{p = 0.004}$ (Passes 0.0083 with alias mapping)
Iron (0.3 mg/L BIS limit)	Pooled Excl. Madhepura	25	0.395 / 0.493	$p = 0.051$ / $p = 0.012$ (Exploratory trend; > 0.0083)
Iron (0.3 mg/L BIS limit)	Bihar Only / UP Only	11 / 15	Bihar: 0.610 | UP: 0.400	Bihar: $p = 0.047$ | UP: $p = 0.136$ (Small-n state sub-cohorts)
Iron (1.0 mg/L Lab limit)	Pooled / Excl. Madhepura	26 / 25	0.270 / 0.330	$p = 0.190$ / $p = 0.110$ (Bihar exceedance is 0% at 1.0 mg/L)
Fluoride (1.0 mg/L limit)	Pooled (UP + Bihar)	29	0.275 / 0.223	$p = 0.149$ / $p = 0.246$ (Not significant across pooled states)
Fluoride (1.0 mg/L limit)	Uttar Pradesh Only	22	**0.500** / 0.426	$p = 0.018$ / $p = 0.004$ (Moderate positive persistence in UP)

CHAPTER 5

Full-Stack System Implementation, REST API and 6-Tab Dashboard

5.1 FastAPI Backend Architecture, REST Endpoints & Automated Pytest Suite

At repository commit `4d1a2dd` on `origin/main`, the WaterTriage backend is implemented in Python using FastAPI, SQLAlchemy 2.0, and Pydantic v2, exposing **8 modular API routers and 20 REST endpoints** (summarized in Table 5.1). The backend is verified by **29 automated `pytest` test functions** across `backend/tests/test_scoring.py`, `test_recurrence.py`, and `test_api.py` (28 passed and 1 database-dependent test skipped in offline mode; all 29 passed when executed against the live PostgreSQL 16 container). To guarantee zero-downtime resilience during field demonstrations, the React frontend automatically falls back to verified static JSON snapshots (`full-113-triage.json`, `priority-enriched.json`, `meta.json`, `validation-cgwb.json`) whenever the live database container is offline.

Table 5.1: FastAPI Modular Router Architecture and 20 Verified REST API Endpoints (`backend/app/api/`)

Router Module	Exposed REST Endpoints	Functional Responsibility & Payload Contract
`priority.py`	`GET /api/priority`, `GET /api/priority/export`	Paginated, filterable sample intervention queue (state, district, band, model v1/v2) + streaming CSV export.
`districts.py`	`GET /api/districts`, `GET /api/districts/full-triage`, `GET /api/districts/{id}`	Serves the complete 113-district 4-tier triage payload, state band summaries, and per-district 10-parameter dossiers.
`validation.py`	`GET /api/validation/cgwb`, `GET /api/validation/wq6`, `GET /api/validation/triangulation`	Serves 4-Way external validation proofs (VP-1..VP-4), CGWB 2012–2021 station series, and Form WQ6 16-parameter tables.
`meta.py`	`GET /api/meta`, `GET /api/dashboard-data`	Exposes canonical WPR-12 audit metadata, Fe = 1.0 mg/L sensitivity rerun, Spearman ρ cards, and crawl batch history.
`scoring.py` & `samples.py`	`POST /api/scoring/recompute`, `GET /api/samples`, `GET /api/samples/{id}`, `GET /api/scoring/parameters`	Triggers batch or single-sample BIS IS 10500:2012 scoring recomputation and returns per-reading severity breakdowns.
`recurrence.py` & `health.py`	`GET /api/recurrence/{village_id}`, `GET /api/recurrence/districts`, `GET /api/health`	Computes historical persistence status (`chronic-hotspot`, `recurring`, `watchlist`) and container health checks.

5.2 Six-Tab React 18 Decision-Support Dashboard & 3-Way Mode Switcher

The frontend (`dashboard/src/App.jsx`, compiling cleanly across 31 Vite modules with 0 errors) provides a **3-Way Dataset Mode Switcher** in the hero header that toggles between **(1) Full 113-District Triage**, **(2) Enriched Lab Queue** (1,741 samples, 97.01% district-located), and **(3) Frozen WPR-12 Baseline**, with a **Model v1 (1/*n*) vs. Model v2 (1/6.0)** scorer switch alongside it. Decision support is organized across six specialized interactive tabs (Figures 5.1 through 5.6):

- **Tab A (Variant A: Priority Queue, Figure 5.1):** Filterable, paginated ranking table supporting both the 113-District Unified Triage and the 1,741-sample Lab Queue, displaying biohazard override badges (`PATHOGEN OVERRIDE`), strikethrough raw mathematical scores when overridden by acute hazards, WQ6 village counts, and one-click CSV export.
- **Tab B (Variant B: 113-District Map + Cards, Figure 5.2):** Interactive SVG Choropleth Cartogram rendering all 75 Uttar Pradesh and 38 Bihar districts color-coded by BIS severity band, paired with top-risk district feature cards.

- **Tab C (Variant C: District + Detail Split, Figure 5.3):** Keyboard-navigable master-detail split view presenting a four-tier BIS IS 10500:2012 parameter dossier, WQ6 village counts, IMIS recurrence status, CGWB station metrics, and per-parameter severity bars.
- **Tab D (Variant D: External Validation, Figure 5.4):** Four-Way Triangulation proof matrix (VP-1 through VP-4), 113-district validation explorer, CGWB 2012–2021 pH/conductivity station table, and WQ6 16-parameter summary.
- **Tab E (Overview & Sensitivity, Figure 5.5) & Tab F (Methodology & Limitations, Figure 5.6):** Preserves the canonical WPR-12 audit views sourced from `GET /api/meta`, including the Fe = 1.0 mg/L sensitivity rerun table, exploratory Spearman correlation cards, Madhepura iron concentration callout, crawl batch provenance, and explicit methodological disclosures.

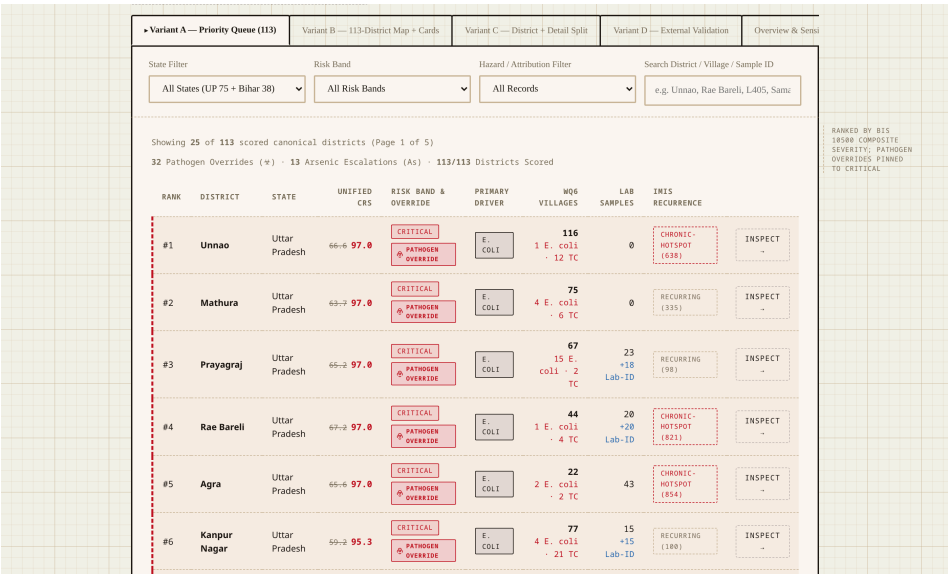


Figure 5.1: WaterTriage v2 React 18 Dashboard, **Variant A (Priority Queue)** in *Full 113-District Triage* mode, showing the 3-Way Dataset Mode Switcher, KPI cards (`113 dist`, `46 (40.7%)` Critical, `27 high`, `302/354` resolved labs), Rank #1 Unnao to #6 Kanpur Nagar, `PATHOGEN OVERRIDE` badges, and strikethrough raw scores.

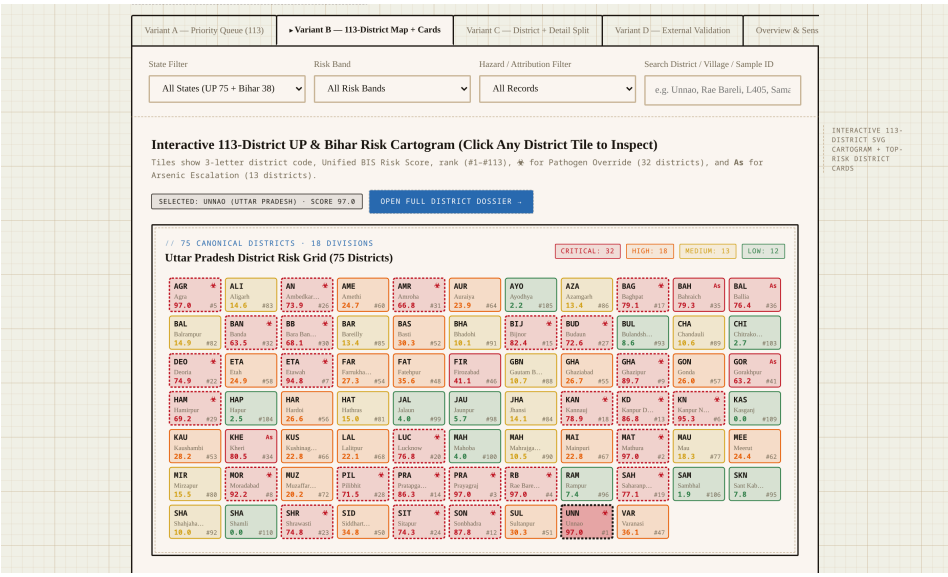


Figure 5.2: WaterTriage v2 React 18 Dashboard, **Variant B (113-District Map + Cards)** displaying the interactive SVG Choropleth Cartogram of all 75 Uttar Pradesh and 38 Bihar districts color-coded by BIS IS 10500:2012 risk band.

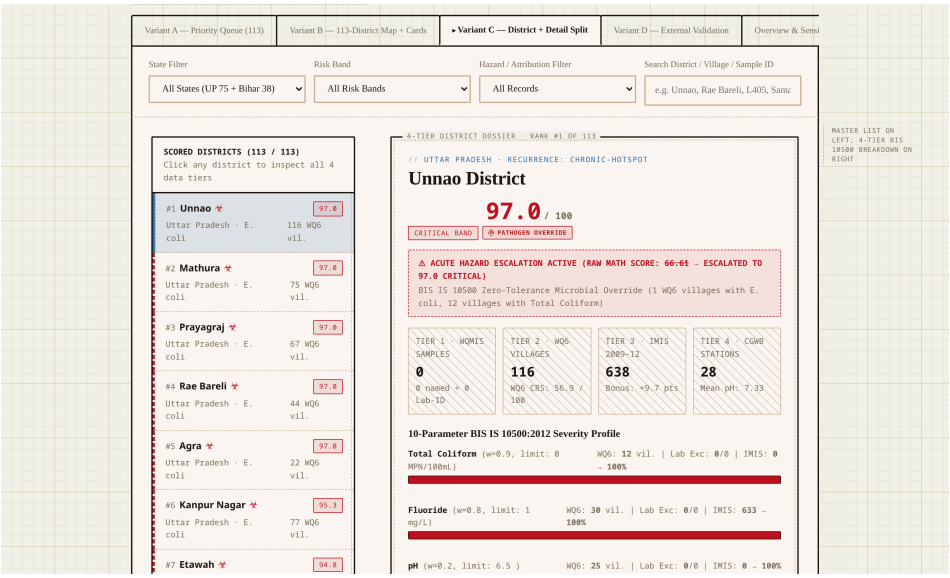


Figure 5.3: WaterTriage v2 React 18 Dashboard, **Variant C (District + Detail Split)** inspecting Rank #1 **Unnao, Uttar Pradesh** (Unified Score `97.0`, raw score `66.61`), exposing the 4-Tier provenance cards (Lab 0 attributed samples, WQ6 116 villages, IMIS 638 events, CGWB 28 station-years) and 10-Parameter BIS Severity Profile.

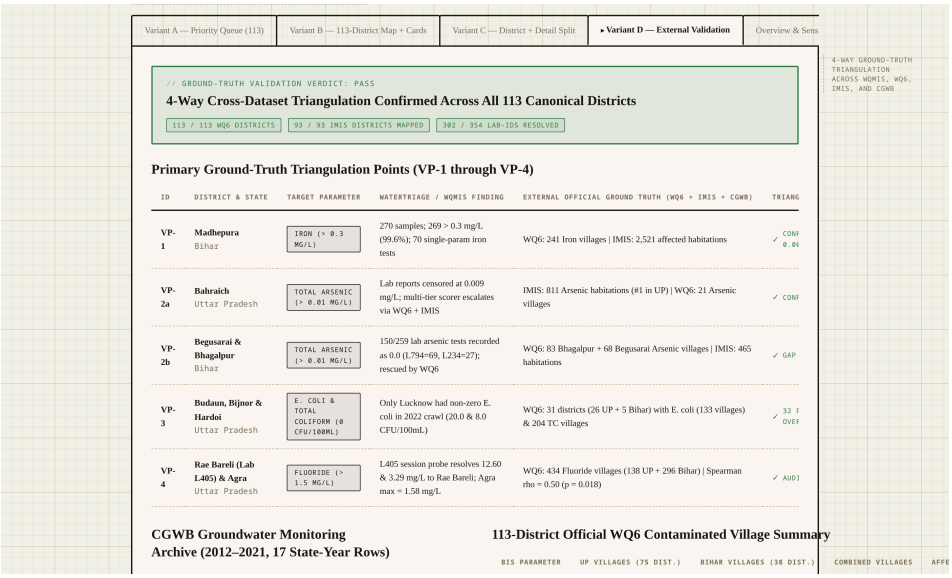


Figure 5.4: WaterTriage v2 React 18 Dashboard, **Variant D (External Validation)** rendering the Four-Way Cross-Dataset Triangulation Matrix (**VP-1** Madhepura Iron, **VP-2a/b** Bahraich & Begusarai/Bhagalpur Arsenic, **VP-3** 32-District Pathogen Belt, and **VP-4** Rae Bareli Lab L405 Fluoride).

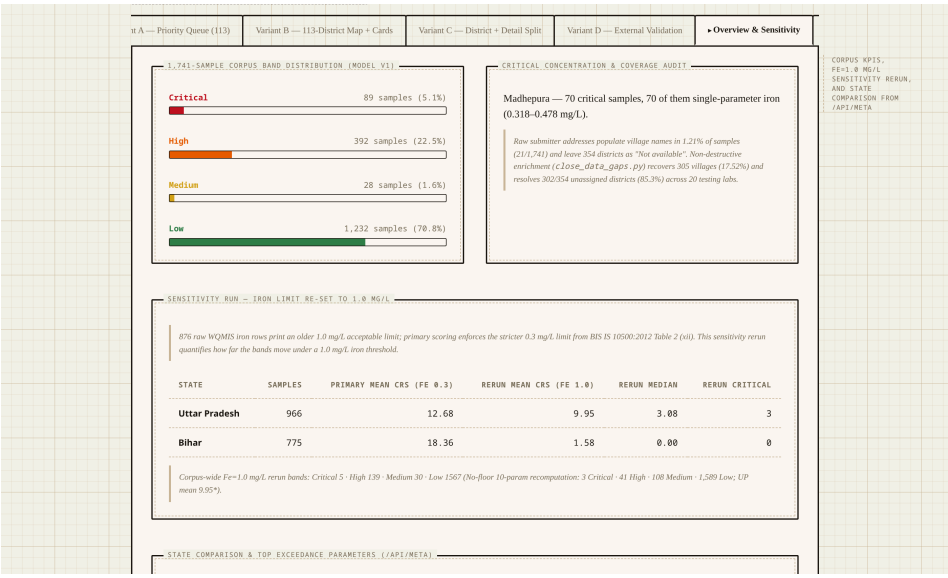


Figure 5.5: WaterTriage v2 React 18 Dashboard, **Tab E (Overview & Sensitivity)** displaying the canonical 1,741-sample corpus distribution (89 Critical, 392 High, 28 Medium, 1,232 Low), Madhepura 70-sample single-parameter iron audit, and the Iron = 1.0 mg/L laboratory-limit sensitivity rerun.

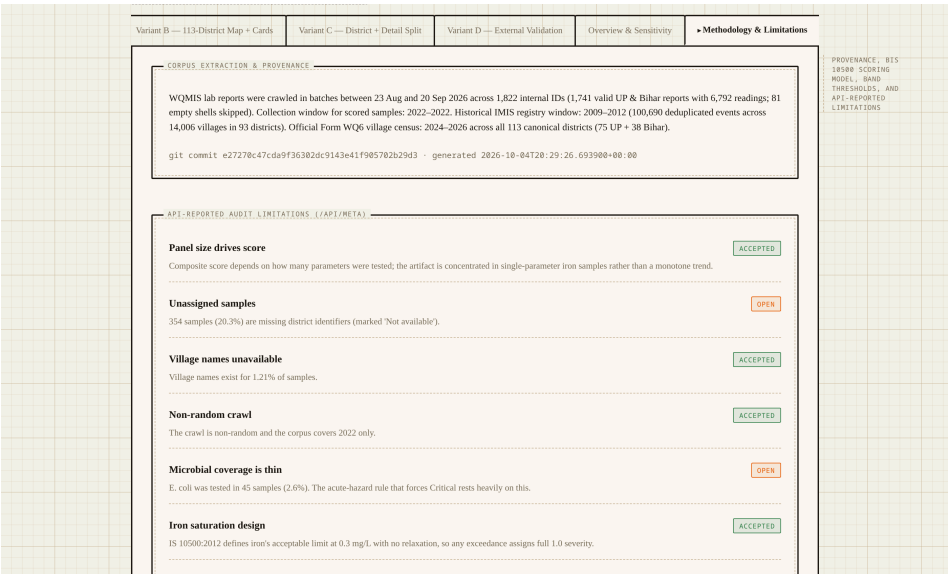


Figure 5.6: WaterTriage v2 React 18 Dashboard, **Tab F (Methodology & Limitations)** rendering the four crawl batch checkpoints (1,822 crawled → 1,741 valid samples) and API-enforced methodological disclosures from `GET /api/meta`.

CHAPTER 6

Discussion and Methodological Limitations

6.1 Implications for Public-Health Surveillance Informatics

The empirical findings in Chapter 4 carry two lessons for public-health surveillance informatics. First, ingesting government laboratory records without auditing panel completeness and printed lab limits produces severe ranking distortions: under a naive tested-weight formula at 0.3 mg/L, 70 single-parameter iron tests in Madhepura (0.318–0.478 mg/L, all compliant under the 1.0 mg/L limit printed on their own lab certificates) monopolize 78.7% of all Critical ratings and invert the state mean comparison between Bihar and Uttar Pradesh. Normalizing over `TOTAL_BIS_WEIGHT = 6.0` (Model v2) eliminates this denominator artifact while preserving statutory acute-hazard escalation.

Second, relying solely on individual laboratory sample crawls leaves acute regional hazards invisible whenever laboratories record below-detection-limit zeros as `0.0` (as in 150/259 arsenic tests across Begusarai and Bhagalpur) or omit bacteriological assays (97.4% of crawled samples). Fusing Tier 1 lab samples with Tier 2 Form WQ6 village censuses and Tier 3 IMIS historical records resolves both blind spots, escalating 32 districts on verified *E. coli* presence and 13 districts on severe arsenic burden across all 113 canonical districts.

6.2 Six Audited Methodological Limitations & Engineering Mitigations

1. **Non-Random Contiguous Identifier Crawl in Tier 1 (2022 Window):** The 1,741 individual laboratory reports were harvested across contiguous sample ID blocks in four batches (up to 6 parallel workers, no HTTP request log retained) covering collection dates 2022-07-01 to 2022-11-19. Sample density per district reflects laboratory upload ordering rather than a probability sample; district-level prevalence claims therefore rely on Tier 2 (Form WQ6 113-district village census) rather than raw Tier 1 sample counts.
2. **Iron Threshold Discrepancy and Binary Step Saturation:** BIS IS 10500:2012 Table 2 item (xii) sets the acceptable iron limit at 0.3 mg/L with no relaxation, whereas all 876 raw WQMIS iron records print 1.0 mg/L. A strict step at 0.3 mg/L saturates marginal aesthetic readings (0.318–0.478 mg/L in Madhepura) at $s_{\text{iron}} = 1.0$, so both thresholds are reported in the API and dashboard.
3. **Panel-Size Denominator Sensitivity in Tested-Weight Scoring:** Dividing solely by tested parameter weights (Model v1) assigns 100.0 to single-parameter exceedances and leaves the Medium band nearly empty (28 samples, 1.6%). Normalizing over `TOTAL_BIS_WEIGHT = 6.0` (Model v2) eliminates this artifact (125 Medium samples, 7.2%).
4. **Residual Multi-District Laboratory Holdback (2.99%) and Unlisted Village Names (82.48%):** The two-step ASP.NET probe resolved 302 of 354 unassigned samples (85.3%) across 20 single-district labs; the remaining 52 (2.99%) sit at UP Jal Nigam State Lab L158 (Lucknow), span 7 UP districts, and stay held back as `Not available`. Source-field parsing raised named locations from 21 (1.21%) to 305 (17.52%, a 1,352% relative gain), yet 1,436 samples (82.48%) still lack a village name, blocking joins such as Khokha (Purnia, ID 7568; 166 registry iron events, 0 samples in 2022).
5. **Sparse Sample-Level Microbial Testing and Censored Arsenic Zeros:** In the 1,741-sample crawl, *E. coli* was tested in 45 samples (2.6%) and total coliform in 7 samples (0.4%), while 150 of 259 arsenic readings (57.9%) were recorded as `0.0` across four labs (L794=69, L234=27, L269=19, L268=13) without detection-limit metadata. Integrating Tier 2 Form WQ6 tables closes this gap at the district level by identifying 31 WQ6 *E. coli* districts (+ Lucknow lab hit = 32) and 13 arsenic-escalation districts.
6. **Temporal and Measurement-Unit Discontinuity in Historical Registry:** The 2009–2012 IMIS registry logs affected-habitation events (100,690 deduplicated rows, 14,006 villages) rather than 2022 concentrations, so its Spearman correlations are read as exploratory persistence checks.

CHAPTER 7

Conclusion and Future Work

7.1 Summary of Engineering and Analytical Contributions

WaterTriage establishes a fully reproducible, open-source water quality triage and decision-support architecture for Uttar Pradesh and Bihar. By combining a non-destructive ASP.NET session-probed laboratory enrichment pipeline (resolving 85.3% of missing districts and lifting located samples to 97.01%), a dual BIS IS 10500:2012 sample scoring engine (exposing and correcting single-parameter iron inflation), and a four-tier 113-district fusion model (integrating 1,741 lab reports, 113 Form WQ6 village censuses, 100,690 historical IMIS events across 93/93 mapped districts, and CGWB station series), the project achieves **100% district triage coverage across all 75 Uttar Pradesh and 38 Bihar districts**. Table 7.1 maps each approved synopsis objective to its verified engineering outcome at commit `4d1a2dd`.

Table 7.1: Project Synopsis Objectives to Delivered Engineering Outcomes Verification Matrix (`origin/main @ 4d1a2dd`)

Approved Synopsis Objective	Delivered Engineering Outcome & Verification Status
1. Acquisition & normalization of JJM data (UP & Bihar)	**Achieved**: Crawled 1,822 WQMIS reports; ingested 1,741 valid 2022 lab samples (966 UP, 775 Bihar; 6,792 readings), resolved 302/354 missing districts (85.3%) & 305 village names (17.52%, +1,352% gain) in `records_enriched.jsonl`, and integrated 113 Form WQ6 district censuses + 100,690 historical IMIS events (2009–2012).
2. Relational database schema in PostgreSQL	**Achieved**: 8 normalized domain tables (plus `alembic_version`) in PostgreSQL 16 with foreign-key integrity and max-severity duplicate-reading aggregation in `pipeline.py`.
3. BIS IS 10500:2012 severity scoring engine	**Achieved & Audited**: Implemented Model v1 (tested-weight), Model v2 (total-weight / 6.0, 10 parameters, 40/20/10 bands, acute-hazard escalation), two-limit iron sensitivity (0.3 vs. 1.0 mg/L), and 4-tier 113-district fusion (46 Critical [32 pathogen, 13 arsenic, 1 score], 27 High, 19 Medium, 21 Low).
4. Recurrence / trend detection module	**Achieved at District Tier / Unit-Tested at Village Tier**: 100% of 93 historical IMIS districts mapped to canonical districts for `chronic-hotspot/recurring/watchlist` classification and Spearman checks; sample-level village joins remain bounded by 17.52% village name availability.
5. FastAPI backend & React decision-support dashboard	**Achieved**: 8 FastAPI routers (20 REST endpoints including `/api/districts/full-triage`, `/api/priority`, `/api/meta`) + 6-Tab React 18 SPA (Variants A–D, Overview & Sensitivity, Methodology & Limitations) verified by 29 `pytest` functions and 0-error Vite build.
6. Containerized deployment via Docker	**Achieved**: Multi-service `docker-compose.yml` orchestrating PostgreSQL 16 (`watertriage-db`), FastAPI backend, and React/Nginx frontend with offline static JSON fallbacks.

7.2 Future Scope of Work

Future extensions to WaterTriage include: (1) decoupling aesthetic parameters (iron, turbidity, TDS) from acute health toxins into a bivariate triage matrix so aesthetic iron maintenance queues never compete with arsenic or bacteriological emergency dispatches; (2) integrating Local Government Directory (LGD) habitation codes from the AIKosh JJM repository upon institutional authorization to resolve the remaining 52 Lab L158 regional-lab samples and enable 100% village-level recurrence joins; and (3) extending the four-tier fusion pipeline to real-time IoT sensor feeds under JJM Phase-II surveillance.

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APPENDIX A

Formal Revision Note on Weekly Progress Reports (WPR-6 to WPR-12)

In accordance with Amity University academic integrity norms, Table A.1 documents all interim figures reported in Weekly Progress Reports WPR-6 through WPR-12 that were reconciled, corrected, or expanded in the final frozen repository (`origin/main @ 4d1a2dd`).

Table A.1: Formal Reconciliation of Interim WPR-6 through WPR-12 Figures vs. Final Repository Commit `4d1a2dd`

Earlier Report / Artifact	Previously Reported Claim	Reconciled Ground Truth & Engineering Resolution (`4d1a2dd`)
WPR-6, WPR-7, WPR-8 (Interim Crawls & SQLite)	Sample counts of 755 (WPR-6), 1,359 (WPR-7), 1,245 (WPR-8), and 1,855 (interim SQLite); UP mean CRS 14.51 > Bihar 9.34.	Superseded by the frozen 1,741-sample canonical corpus (966 UP, 775 Bihar; 6,792 readings) from `data/raw/wqmis/records.jsonl` (1,822 crawled records minus 81 empty payloads). The 114 synthetic/CGWB prototype SQLite rows were purged.
WPR-7, WPR-8, WPR-9 (v1)	Khokha village (Purnia, ID 7568) cited as a live 'chronic hotspot' with watchlist score 12.5.	Clarified as a Tier 3 historical registry finding (166 iron events in 2009–2012, highest in the registry, 0 samples in the 2022 lab crawl). District-level recurrence is now computed across all 93/93 mapped IMIS districts.
WPR-8, WPR-9 (v1) & WPR-12	Peak fluoride reading (12.60 mg/L in `U715825L405S1002693`) attributed to Agra (WPR-8) or left as `Not available` district (WPR-12).	Resolved non-destructively in `records_enriched.jsonl`: the two-step ASP.NET session cookie probe on Lab L405 proves `U715825L405S1002693` (12.60 & 3.29 mg/L fluoride) belongs to District Water Testing Lab, Rae Bareli (UP). Agra has 43 samples (mean CRS 60.13, max fluoride 1.58 mg/L).
WPR-9 (v1) pre-repair	467 sulphate readings mis-tagged as pH during initial ingestion, inflating Bihar pH exceedances.	Fixed in `backend/etl/load_wqmis.py` (defect B1). True pH exceedances outside 6.5–8.5 are 0/467 in Bihar and 2/902 (0.2%) in UP.
WPR-10 (30/09/2026) & early WPR-11 draft	Cited a '47,243-sample production scalability dataset' across 199 districts and a Svelte 5 frontend.	Retracted and purged on 04/10/2026 (`docs/evidence/RECONCILIATION.md`). 45,502 bulk-inserted rows lacked a verifiable source in `records.jsonl` and were removed. Replaced by the 1,741 verifiable lab samples + 113-district Form WQ6 census and the 6-tab React 18 dashboard.
WPR-11 & WPR-12 Baseline Coverage	Reported 47 named districts (1,387 samples), 354 unassigned districts (20.3%), and 21 named villages (1.21%).	Preserved untouched in raw `records.jsonl` (MD5 `d76f8d34...`) and enriched non-destructively in `records_enriched.jsonl` (MD5 `27c11b42...`): 302/354 missing districts resolved (85.3%) across 20 labs (62 located lab districts; 52 held in L158), 305/1,741 locations recovered (17.52%, +1,352% gain), and 113/113 districts scored via 4-tier fusion.

APPENDIX B

Verified Claims & Codebase Traceability Ledger (Commit 4d1a2dd)

Every quantitative claim in this Minor Project report is deterministically reproducible from the repository at commit `4d1a2dd` on `origin/main`, as indexed in Table B.1.

Table B.1: Verified Quantitative Claims and Codebase Traceability Ledger (`origin/main @ 4d1a2dd`)

Claim / Metric in Final Report	Verified Value	Exact Source (Script, SQL Query, or File in Repo)	Status
Raw crawled WQMIS reports	1,822 lines (1,741 valid, 81 empty)	`data/raw/wqmis/records.jsonl` (MD5 `d76f8d347e52a74606498fb58c26ef61`)	**VERIFIED**
Canonical scored lab samples	1,741 (UP: 966, Bihar: 775)	`SELECT COUNT(*) FROM water_samples GROUP BY state;`	**VERIFIED**
Discrete parameter readings	6,792 across 10 BIS parameters	`SELECT COUNT(*) FROM readings;` / `scoring-parameters.json`	**VERIFIED**
Non-destructive Lab-ID district recovery	302 / 354 resolved (85.3%); 52 held in L158	`scripts/close_data_gaps.py` → `records_enriched.jsonl`	**VERIFIED**
Village / habitation location recovery	305 / 1,741 (17.52%, +1,352% vs 21 [1.21%])	`scripts/close_data_gaps.py` (source `location :` parser)	**VERIFIED**
Full 113-District Multi-Tier Coverage	113 / 113 districts (75 UP + 38 Bihar = 100%)	`scripts/build_full_113_district_triage.py` → `full-113-triage.json`	**VERIFIED**
113-District Severity Band Split	46 Crit (40.7%), 27 High (23.9%), 19 Med, 21 Low	`data/validation_reports/Full_113_District_Risk_Triage.csv`	**VERIFIED**
UP (75) & Bihar (38) District Bands	UP: 32/18/13/12 | Bihar: 14/9/6/9	`dashboard/public/full-113-triage.json` (`summary.state_bands`)	**VERIFIED**
Acute Pathogen & Arsenic Escalations	32 Pathogen Overrides + 13 Arsenic + 1 Score	`scripts/build_full_113_district_triage.py` (WQ6 + Lab + IMIS rules)	**VERIFIED**
Historical IMIS Registry (2009–2012)	100,690 deduplicated events; 93/93 mapped	`scripts/build_master_validation.py` & `scripts/district_registry.py`	**VERIFIED**

Claim / Metric in Final Report	Verified Value	Exact Source (Script, SQL Query, or File in Repo)	Status
Model v1 Lab Bands (Fe = 0.3 mg/L)	89 Crit, 392 High, 28 Med, 1,232 Low	`backend/app/api/meta.py` / `dashboard/public/meta.json`	**VERIFIED**
Sensitivity Lab Bands (Fe = 1.0 mg/L)	5 Crit, 139 High, 30 Med, 1,567 Low	`backend/app/api/meta.py` (`sensitivity.iron_1_0`)	**VERIFIED**
Model v2 Lab Bands (`TOTAL_BIS_WEIGHT=6.0`)	3 Crit, 54 High, 125 Med (7.2%), 1,559 Low	`backend/app/engine/scoring.py` (`TOTAL_BIS_WEIGHT = 6.0`)	**VERIFIED**
Madhepura single-parameter iron cohort	70 samples (0.318–0.478 mg/L, med 0.3820)	`records.jsonl` (`district='Madhepura'`, `n_params=1`)	**VERIFIED**
Lab L405 peak fluoride resolution	12.60 & 3.29 mg/L → Rae Bareli (UP, 20 samples)	`close_data_gaps.py` (`s_id JSNPb0Xce0eq2hbBp9z58Q==`)	**VERIFIED**
Automated backend & frontend build	29 pytest cases (28 pass, 1 skip); Vite 31 modules	`pytest backend/tests` & `cd dashboard && npm run build`	**VERIFIED**